

NANYANG TECHNOLOGICAL UNIVERSITY
SPMS/DIVISION OF MATHEMATICAL SCIENCES

2013/14 Semester 1

MH1300 Foundations of Mathematics

Tutorial 2

For the tutorial on 26 August, let us discuss

- Ex. 2.1. 22, 24, 33, 42, 52,
- Ex. 2.2. 12, 13a, 39.

You are encouraged to try the following problems. We will not be discussing them, but solutions will be provided.

- Ex. 2.1. 28, 49,
- Ex. 2.2. 20e, 22e, 23e, 40, 45, 49.

Ex. 2.1.22, 24 Determine which of the pairs of statement forms are logically equivalent. Justify your answers using truth tables and include a few words of explanation.

22. $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$.

24. $(p \vee q) \vee (p \wedge r)$ and $(p \vee q) \wedge r$.

[Solution:]

22.

p	q	r	$q \vee r$	$p \wedge q$	$p \wedge r$	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$
1	1	1	1	1	1	1	1
1	1	0	1	1	0	1	1
1	0	1	1	0	1	1	1
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
0	1	0	1	0	0	0	0
0	0	1	1	0	0	0	0
0	0	0	0	0	0	0	0

same truth values

The truth table shows that $p \wedge (q \vee r)$ and $(p \wedge q) \vee (p \wedge r)$ always have the same truth values. Therefore they are logically equivalent. This proves the distributive law for $\wedge$ over $\vee$.

24.

p	q	r	$p \vee q$	$p \wedge r$	$(p \vee q) \vee (p \wedge r)$	$(p \vee q) \wedge r$
1	1	1	1	1	1	1
1	1	0	1	0	1	0
1	0	1	1	1	1	1
1	0	0	1	0	1	0
0	1	1	1	0	1	1
0	1	0	1	0	1	0
0	0	1	0	0	0	0
0	0	0	0	0	0	0

different truth values

The truth table shows that $(p \vee q) \vee (p \wedge r)$ and $(p \vee q) \wedge r$ have different truth values in rows 2, 4, and 6. Hence they are not logically equivalent.

Ex. 2.1.28. Use De Morgan's laws to write a negation for the statement

"This computer program has a logical error in the first ten lines or it is being run with an incomplete data set."

[Solution.]

This computer program does not have a logical error in the first ten lines and it is not being run with an incomplete data set.

Note the statement "This computer program does not have a logical error in the first ten lines and it is being run with a complete data set." is not an acceptable answer. The negation of "it is being run with an incomplete data set" is not "it is being run with a complete data set". For example, it could be the case that the program is not running at all.

Ex. 2.1.33. Assume x is a particular real number and use De Morgan's laws to write negations for the statement

$$-10 < x < 2.$$

[Solution.]

$$-10 \geq x \quad \text{or} \quad x \geq 2.$$

(Here we are assuming $x \in \mathbb{R}$.)

Ex. 2.1.42. Use truth tables to establish which of the statement forms are tautologies and which are contradictions.

$$((\neg p \wedge q) \wedge (q \wedge r)) \wedge \neg q.$$

[Solution.]

p	q	r	$\neg p$	$\neg p \wedge q$	$q \wedge r$	$\neg q$	$(\neg p \wedge q) \wedge (q \wedge r)$	$((\neg p \wedge q) \wedge (q \wedge r)) \wedge \neg q$
1	1	1	0	0	1	0	0	0
1	1	0	0	0	0	0	0	0
1	0	1	0	0	0	1	0	0
1	0	0	0	0	0	1	0	0
0	1	1	1	1	1	0	1	0
0	1	0	1	1	0	0	0	0
0	0	1	1	0	0	1	0	0
0	0	0	1	0	0	1	0	0

Since all the truth values of $((\neg p \wedge q) \wedge (q \wedge r)) \wedge \neg q$ are 0's, $((\neg p \wedge q) \wedge (q \wedge r)) \wedge \neg q$ is a contradiction.

Ex. 2.1.49. A logical equivalence is derived from Theorem 2.1.1. Supply a reason for each step.

$$\begin{aligned}
 (p \vee \neg q) \wedge (\neg p \vee \neg q) &\equiv (\neg q \vee p) \wedge (\neg q \vee \neg p) && \text{by (a)} \\
 &\equiv \neg q \vee (p \wedge \neg p) && \text{by (b)} \\
 &\equiv \neg q \vee \mathbf{F} && \text{by (c)} \\
 &\equiv \neg q && \text{by (d)}
 \end{aligned}$$

Therefore, $(p \vee \neg q) \wedge (\neg p \vee \neg q) \equiv \neg q$.

[Solution.]

$$\begin{aligned}
 (p \vee \neg q) \wedge (\neg p \vee \neg q) &\equiv (\neg q \vee p) \wedge (\neg q \vee \neg p) && \text{by commutative law} \\
 &\equiv \neg q \vee (p \wedge \neg p) && \text{by distributive law} \\
 &\equiv \neg q \vee \mathbf{F} && \text{by negation law} \\
 &\equiv \neg q && \text{by identity law.}
 \end{aligned}$$

Ex. 2.1.52. Use Theorem 2.1.1 to verify the logical equivalence

$$\neg(p \vee \neg q) \vee (\neg p \wedge \neg q) \equiv \neg p.$$

Supply a reason for each step.

[Solution.]

$$\begin{aligned}
 \neg(p \vee \neg q) \vee (\neg p \wedge \neg q) &\equiv (\neg p \wedge \neg(\neg q)) \vee (\neg p \wedge \neg q) && \text{by De Morgan's law} \\
 &\equiv (\neg p \wedge q) \vee (\neg p \wedge \neg q) && \text{by double negation} \\
 &\equiv \neg p \wedge (q \vee \neg q) && \text{by distributive law} \\
 &\equiv \neg p \wedge \mathbf{T} && \text{by negation law} \\
 &\equiv \neg p && \text{by identity law.}
 \end{aligned}$$

Ex. 2.2.12 (Modified). Apply Ex. 2.2.13a (below) to establish the logical equivalence

$$p \vee q \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r).$$

Next, use it rewrite the following statement. (Assume that x represents a fixed real number.)

$$\text{If } x > 2 \text{ or } x < -2, \text{ then } x^2 > 4.$$

[Solution.]

(a)

$$\begin{aligned}
 p \vee q \rightarrow r &\equiv \neg(p \vee q) \vee r && \text{by Ex. 2.2.13a} \\
 &\equiv (\neg p \wedge \neg q) \vee r && \text{by De Morgan's law} \\
 &\equiv r \vee (\neg p \wedge \neg q) && \text{by commutativity law} \\
 &\equiv (r \vee \neg p) \wedge (r \vee \neg q) && \text{by distributive law} \\
 &\equiv (\neg p \vee r) \wedge (\neg q \vee r) && \text{by commutativity law} \\
 &\equiv (p \rightarrow r) \wedge (q \rightarrow r) && \text{by Ex. 2.2.13a}
 \end{aligned}$$

(b) If $x > 2$ then $x^2 > 4$, and if $x < -2$ then $x^2 > 4$.

Ex. 2.2.13a. Use a truth table to verify the following logical equivalence. Include a few words of explanation with your answer.

$$p \rightarrow q \equiv \neg p \vee q.$$

[Solution.]

p	q	$p \rightarrow q$	$\neg p \vee q$
1	1	1	1
1	0	0	0
0	1	1	1
0	0	1	1

same truth values

The table shows that $p \rightarrow q$ and $\neg p \vee q$ have the same truth values. Therefore, they are logically equivalent.

Ex. 2.2.20e, 22e, 23e. Write the negation, contrapositive, converse, and inverse for the following statement. (Assume that all variables represent fixed quantities or entities, as appropriate.)

“If x is nonnegative, then x is positive or x is 0.”

[Solution.]

Negation: x is nonnegative and x is not positive and x is not 0.

Contrapositive: If x is not positive and x is not 0, then x is negative. (Or “... x is not nonnegative.”)

Converse: If x is positive or x is 0, then x is nonnegative.

Inverse: If x is negative, then x is not positive and x is not 0. (Or “If x is not nonnegative ...”.)

Ex. 2.2.39. Definition: If r and s are statements, r **unless** s means if $\neg s$ then r .

Rewrite the statement in if-then form. “This door will not open unless a security code is entered.”

[Solution:]

If a security code is not entered, then this door will not open.

Ex. 2.2.40. Rewrite the statement in if-then form. “Catching the 8:05 bus is a sufficient condition for my being on time for work.”

[Solution.]

If I catch the 8:05 bus, then I am on time for work.

Ex. 2.2.45. Note that “a sufficient condition for s is r ” means r is a sufficient condition for s and that “a necessary condition for s is r ” means r is a necessary condition for s . Rewrite the following statement in if-then form.

A necessary condition for this computer program to be correct is that it not produce error messages during translation.

[Solution.]

Let r be “this computer program **does not** produce error messages during translation” and s be “this computer program is correct”.

Then the two acceptable forms are “if $\neg r$ then $\neg s$ ” and “if s then r ”.

If this computer program produces error messages during translation, then it is not correct.

OR

If this computer program is correct, then it does not produce error messages during translation.

Ex. 2.2.49. (a) Use the logical equivalences

$$p \rightarrow q \equiv \neg p \vee q \quad \text{and} \quad p \leftrightarrow q \equiv (\neg p \vee q) \wedge (\neg q \vee p)$$

to rewrite the given statement form without the symbol $\rightarrow$ and $\leftrightarrow$, and (b) use the logical equivalence

$$p \vee q \equiv \neg(\neg p \wedge \neg q)$$

to rewrite the given statement form using only $\wedge$ and $\neg$.

$$\text{Given statement:} \quad (p \rightarrow r) \leftrightarrow (q \rightarrow r).$$

[Solution.]

(a)

$$\begin{aligned} & (p \rightarrow r) \leftrightarrow (q \rightarrow r) \\ & \equiv (\neg p \vee r) \leftrightarrow (\neg q \vee r) && \text{(by first logical equivalence in (a))} \\ & \equiv [\neg(\neg p \vee r) \vee (\neg q \vee r)] \wedge [\neg(\neg q \vee r) \vee (\neg p \vee r)] && \text{(by second logical equivalence in (a))} \\ & && \text{(an acceptable answer)} \\ & \equiv [(p \wedge \neg r) \vee (\neg q \vee r)] \wedge [(q \wedge \neg r) \vee (\neg p \vee r)] && \text{(by De Morgan's law)} \\ & && \text{(another acceptable answer)} \end{aligned}$$

(b) Continuing from Part (a),

$$\begin{aligned} & \equiv [(p \wedge \neg r) \vee \neg(\neg\{\neg q\} \wedge \neg r)] \wedge [(q \wedge \neg r) \vee \neg(\neg\{\neg p\} \wedge \neg r)] && \text{(by logical equivalence in (b))} \\ & \equiv [(p \wedge \neg r) \vee \neg(q \wedge \neg r)] \wedge [(q \wedge \neg r) \vee \neg(p \wedge \neg r)] && \text{(by double negation)} \\ & \equiv \neg[\neg(p \wedge \neg r) \wedge \neg\{\neg(q \wedge \neg r)\}] \wedge \neg[\neg(q \wedge \neg r) \wedge \neg\{\neg(p \wedge \neg r)\}] && \text{(by logical equivalence in (b))} \\ & \equiv \neg[\neg(p \wedge \neg r) \wedge (q \wedge \neg r)] \wedge \neg[\neg(q \wedge \neg r) \wedge (p \wedge \neg r)] && \text{(by double negation)} \end{aligned}$$